

June 22, 2026

Editor-in-Chief
Sensors
MDPI

Dear Editor,

I am pleased to submit our manuscript entitled “A Real-Time Visual Sensing System for Classroom Engagement Monitoring Using Multi-Scale Temporal Neural Networks” for consideration for publication in *Sensors*.

This paper presents a camera-based visual sensing system for automated classroom engagement monitoring. The system processes video streams from standard classroom cameras using a lightweight neural network architecture, providing real-time engagement detection without requiring specialized hardware or wearable sensors.

Key contributions relevant to *Sensors*:

1. **Visual sensing system design:** A complete pipeline from camera input to engagement classification, optimized for real-time classroom deployment.
2. **Multi-scale attention:** Parallel 3×3 , 5×5 , and 7×7 convolutions with adaptive fusion for robust facial expression analysis across different scales.
3. **Temporal aggregation:** BiLSTM-based video-level prediction achieving 94.8% accuracy on DAiSEE, demonstrating the importance of temporal analysis in visual sensing systems.
4. **Edge deployment:** 1.26M parameters, 53 fps on standard GPU, suitable for resource-constrained classroom environments.

The system is evaluated on three benchmark datasets (FER2013, CK+, DAiSEE) with comprehensive ablation studies. Results demonstrate that camera-based visual sensing can provide practical, non-intrusive engagement monitoring for educational environments.

Both authors have read and approved the manuscript. We declare no conflicts of interest.

I look forward to your decision.

Sincerely,

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